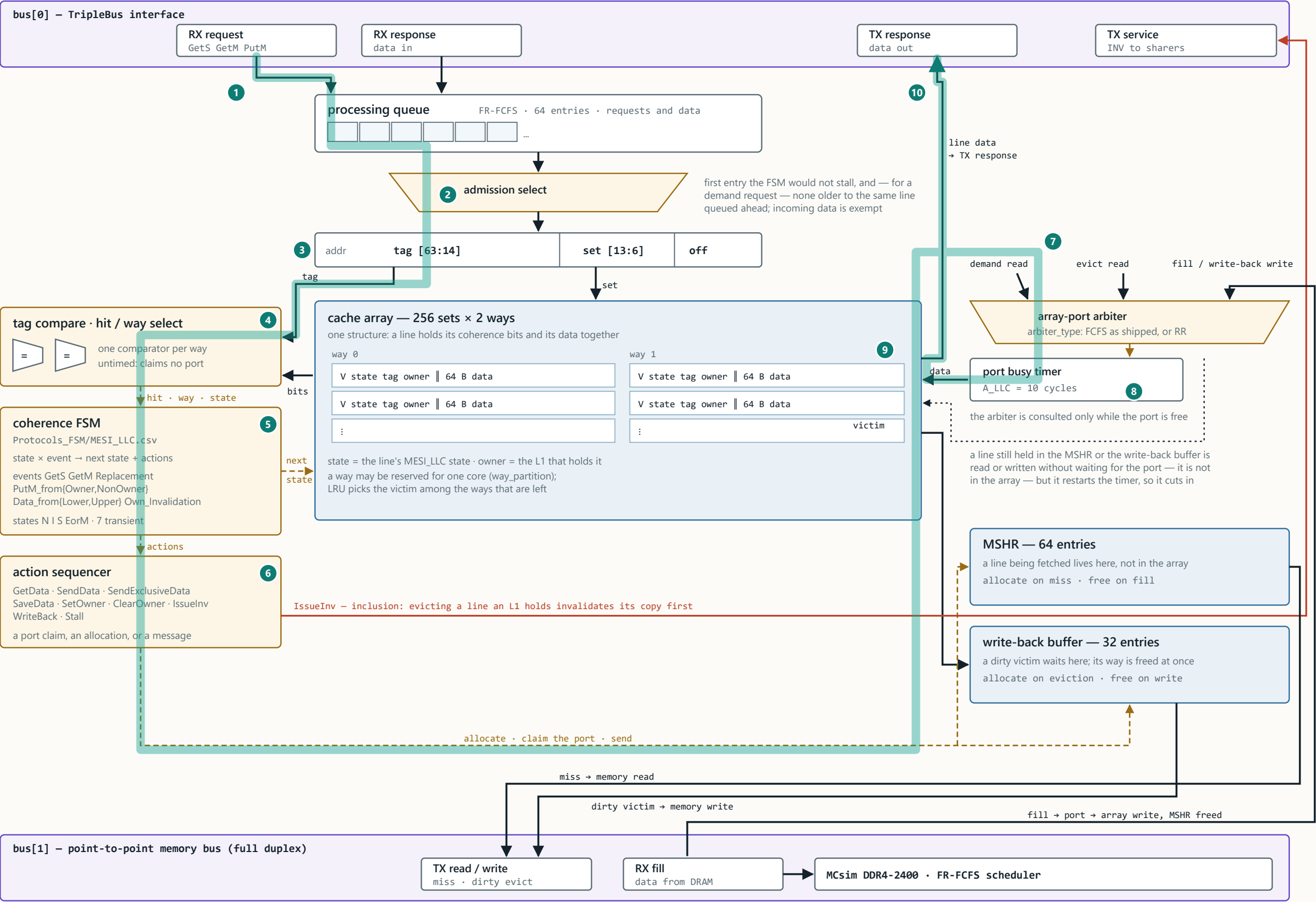


Octopus LLC datapath — the path of a read / write hit

CacheController_End2End + CacheDataHandler_COTS · 32 KB, 2-way, 64 B lines · MESI_LLC coherence FSM · one timed data port

Highlighted: a GetS / GetM for a line the LLC holds valid and no L1 owns. A write hit takes the same path and also rewrites the owner bits (untimed); sharers invalidate themselves by snooping the GetM on the bus — the LLC sends no INV (red is eviction only).



— data path (timed) register-resident access, bypasses the port - - - - - control (untimed) — back-invalidation (inclusion)

□ storage □ control logic □ interconnect interface the Logger and the raw trace observe: queue arrival · admission · port claim · data emitted

— read / write hit — 1 arrives · 2 admitted · 3 decoded · 4 tag hit (untimed) · 5 FSM: send data, exclusive if unshared; set owner unless a GetS finds it shared · 6 sequencer claims the port · 7 arbiter grants · 8 busy timer · 9 array read · 10 data out on TX response